

Ritul Rashi

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EDUCATION

Virginia Tech

Blacksburg, VA

Bachelor of Science in Computer Science

Expected Graduation: May 2026

Relevant Coursework: Software Design and Data Structures, Computer Organization, Discrete Mathematics, Statistics, Applied Combinatorics, Agile/Scrum Development, Mobile App Development (Swift), Computer Systems

Clubs: Association for Women in Computing at Virginia Tech

SKILLS

Languages: Java, Python, Go, C++, C, SQL, JavaScript, TypeScript

Backend & Systems: Spring Boot, FastAPI, Node.js, Kafka, REST APIs

Cloud & Infra: AWS (EC2, S3, Lambda), Kubernetes, Docker, Linux

Security & QA: RBAC, OAuth2, JWT, API Testing (Postman)

Distributed Systems & Data Integrity: Transaction Processing, Reconciliation, Data Integrity, KPI Reporting

PROJECTS

Personal Web & Video Server, Concurrent HTTP/1.1 Server — Virginia Tech CS3214

- Built a concurrent HTTP/1.1 web server in C supporting persistent connections, video streaming, and IPv4/IPv6 networking.
- Implemented stateless JWT authentication with HMAC signing, secure cookie management, and token expiration controls.
- Designed a multi-threaded client handler supporting connections, HTML5 fallback routing, and malformed-request resilience.

InsightOps—Distributed System Observability Platform

- Built a Kubernetes-based observability stack using Prometheus, Loki, and Helm for real-time metrics and logging
- Developed a Go-based anomaly detection service integrated with FastAPI to identify system failures and performance degradation
- Automated failure classification and alerting pipelines, improving incident response and reducing debugging time insights.

Midas Core—Distributed Transaction Processing System

- Built a distributed transaction platform using 7 Spring Boot microservices, Apache Kafka, and PostgreSQL for financial processing.
- Designed a double-entry ledger with automated reconciliation services and scheduled integrity checks for transactional accuracy.
- Developed a centralized JWT/RBAC API Gateway and secure REST APIs with strict request validation controls.
- Implemented idempotent event processing and 13 integration tests to ensure system reliability and consistency.

UPICrypto—Decentralized Payment Infrastructure (Blockchain-based)

- Developed 6 Solidity smart contracts on Ethereum and Polygon for decentralized payment processing and asset transfers.
- Implemented EIP-712 typed-signature workflows and gas-optimized contract logic for secure transaction authorization.
- Created 84 automated Hardhat tests covering contract functionality, edge cases, and security-critical workflows.
- Designed blockchain infrastructure prioritizing security, auditability, and integrity for decentralized financial transactions.

WORK EXPERIENCE

Research & Backend Engineer, DBWorkout Platform — Virginia Tech

January 2025 - Present

- Designed a PostgreSQL-backed sandbox system supporting concurrent query execution and real-time grading for 170+ users
- Implemented secure authentication and authorization using JWT, bcrypt, and CAS, enforcing strict RBAC policies across roles
- Built and executed automated API validation suites using Postman, ensuring payload integrity and access control correctness
- Strengthened system reliability by identifying and resolving failure points in backend request handling and query execution.

Full-Stack Developer — PSN Musicals

June 2024 - August 2024

- Architected and deployed a multi-service platform on AWS (EC2, S3, Lambda) with containerized services and scalable APIs
- Optimized SQL queries and backend request handling, reducing latency by 25% under concurrent workloads
- Implemented OAuth2 authentication and RBAC, enforcing least-privilege access across distributed components
- Delivered production features using Agile workflows with zero-downtime deployments and stable release cycles.

Software Developer Intern — DuoFits

May 2023 - August 2023

- Optimized SQL pipelines and indexing strategies, significantly improving query performance under production load
- Built automated reporting systems generating real-time KPIs and structured analytics for internal stakeholders
- Partnered with engineering and QA teams to design testing workflows and validate system behavior across deployments.

CERTIFICATIONS

Options 101 Course — Akuna Capital

August 2025

AWS Solutions Architect — AWS

August 2023

Machine Learning — IIT Allahabad

August 2024